

Vinci — Final Project Catalog

CS6460 Educational Technology | Allan Kangethe | akangethe3@gatech.edu

About

Vinci is a Flask web application that implements a three-stage scaffolded GenAI feedback system for novice Python programmers. Learners work through Socratic questioning, constrained code review via unit tests, and a worked-example fallback before seeing a full solution. This is a development track project. The tool runs locally.

How to Run

Use this link for the live demo:

<https://akanget1.pythonanywhere.com/login>

External Links

GitHub: <https://github.com/Amkangethe/final-project>

Presentation Video: <https://akanget1.pythonanywhere.com/login>

Contents

Catalog.pdf — This file.

README.md — Setup instructions and project overview.

app.py — Main Flask app with all five API endpoints.

ollama_ai.py — OllamaService class — handles LLM interaction and stage-label prompting. **requirements.txt** — Python dependencies.

sessions_log.json — Pilot session data (one completed dedupe session).

templates/ — HTML templates — index.html (main UI) and login.html.

static/ — CSS, JavaScript, sample challenge files, and images.

tests/ — pytest unit test files for all six Python challenges.

Project_Proposal.pdf — Original project proposal with design rationale and evaluation plan.